



***ESE 5760: Semiconductor Memory Devices
And Circuit Design***

Final Project Proposal

**A Pythonic Refinement of DESTINY for Emerging
Memory Tradeoff Exploration**

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1. Project Proposal

In this project, our team plans to use the DESTINY simulator and the upcoming Python-based port of this modeling framework to investigate trade-offs in emerging memory technologies. DESTINY is a design space exploration tool for 2D and 3D cache architectures that supports SRAM, eDRAM, PCM, STT-RAM, and ReRAM. It also allows designers to investigate architectural parameters such as the number of stacked chips, partition granularity, TSV projection assumptions, and different combinations of tag/data technologies. These features make it a powerful starting point for exploring emerging memory systems.

This project will focus on conducting an in-depth technical evaluation of DESTINY for a selected emerging memory design (a 3D cache with SRAM as the tag array). We have chosen this direction because DESTINY explicitly supports heterogeneous tag/data technologies and various stacked-die configurations, making it suitable for comparing trade-offs between density, latency, power consumption, and area in real-world cache design scenarios. Furthermore, the tool supports the import of multiple memory cell input files and can generate optimal design solutions for different optimization objectives, which will enable us to systematically compare the results of baseline and improved designs.

Our methodology will consist of four steps. First, we will use the original DESTINY tool to reproduce a benchmark design study to understand its configuration process, output metrics, and optimization behavior. Second, we will examine the code structure and modeling assumptions to identify potential flaws, such as oversimplified treatment of 3D TSV overhead, rigid partitioning assumptions, or inaccuracies in modeling emerging memory behavior. Third, we will implement a series of targeted fixes for the Python port, prioritizing improvements that can be quantitatively evaluated. Finally, we will conduct comparative experiments between the original and improved versions of the model using the same memory configuration space, measuring how these changes affect conclusions regarding area, latency, power consumption, and overall design optimality.

The expected outcome of this project is to establish a more transparent and accurate modeling workflow for emerging memory technologies. In addition to porting the simulator to Python, our goal is to demonstrate that the assumptions of the improved model can substantially alter design conclusions, particularly when comparing different 3D architectures and emerging memory technologies.

2. Project Time Line

The project will be completed over eight weeks, beginning with background study and baseline tool reproduction, followed by flaw identification, Python-based model refinement, quantitative evaluation, and final report preparation.

Late March: Team formation, project scoping, literature review, and target memory design selection.

Late March to early April: Baseline DESTINY simulations.

Early April to late April: Code inspection, development of a Pythonic porting framework, and implementation of model improvements.

Late April to early May: Quantitative evaluation and result analysis.

Early May to mid-May: Report drafting, revision, and final submission.